ECA0801 - ANALOG AND DIGITAL COMMUNICATION

Exp7: CALCULATION AND VERIFICATION OF FREQUENCY DEVIATION AND MODULATION INDEX OF FM SIGNALS USING MATLAB SIMULATION

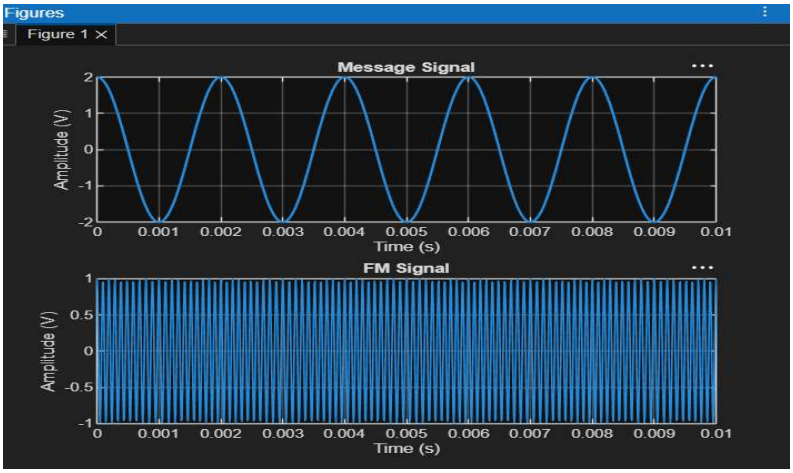
Aim:

To generate a Frequency Modulated (FM) signal in MATLAB using specified carrier and message signal parameters.

2. To calculate and verify the frequency deviation of the FM signal from the given carrier frequency and modulating signal parameters.

3. To determine and verify the modulation index of the FM signal using the relationship between frequency deviation and message frequency.

4. To analyze the effect of message frequency on frequency deviation and modulation index, and verify their relationship through MATLAB simulation.



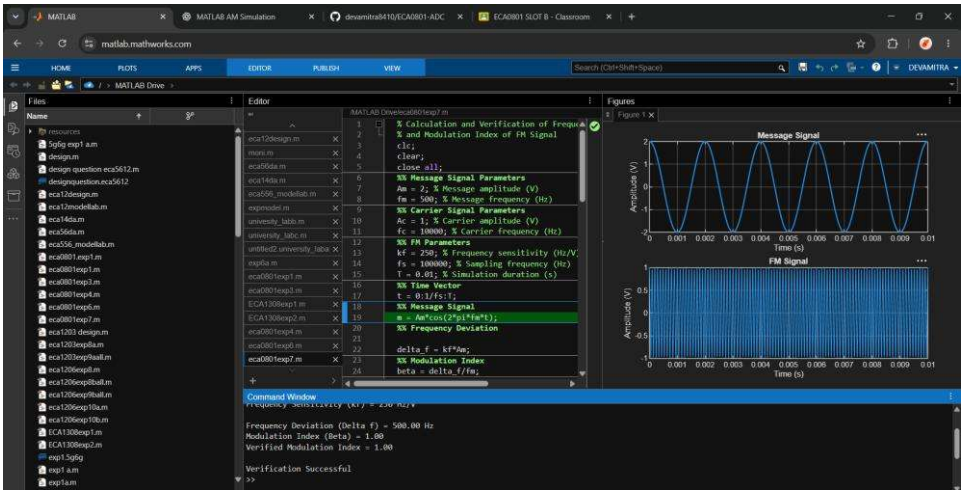
Program:

```
% Calculation and Verification of Frequency Deviation
% and Modulation Index of FM Signal
clc;
clear;
close all;
%% Message Signal Parameters
Am = 2; % Message amplitude (V)
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```
fm = 500; % Message frequency (Hz)
%% Carrier Signal Parameters
Ac = 1; % Carrier amplitude (V)
fc = 10000; % Carrier frequency (Hz)
%% FM Parameters
kf = 250; % Frequency sensitivity (Hz/V)
fs = 100000; % Sampling frequency (Hz)
T = 0.01; % Simulation duration (s)
%% Time Vector
t = 0:1/fs:T;
%% Message Signal
m = Am*cos(2*pi*fm*t);
%% Frequency Deviation
delta_f = kf*Am;
%% Modulation Index
beta = delta_f/fm;
%% FM Signal
fm_signal = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));
%% Display Parameters
fprintf('Carrier Frequency (fc) = %.0f Hz\n',fc);
fprintf('Message Frequency (fm) = %.0f Hz\n',fm);
fprintf('Message Amplitude (Am) = %.0f V\n',Am);
fprintf('Frequency Sensitivity (kf) = %.0f Hz/V\n',kf);
fprintf('\nFrequency Deviation (Delta f) = %.2f Hz\n',delta_f);
fprintf('Modulation Index (Beta) = %.2f\n',beta);
%% Verification
beta_verify = delta_f/fm;
fprintf('Verified Modulation Index = %.2f\n',beta_verify);
%% Verification Check
tolerance = 1e-6;
if abs(beta - beta_verify) < tolerance
fprintf('\nVerification Successful\n');
else
fprintf('\nVerification Failed\n');
end
%% Plotting
figure;
subplot(2,1,1);
plot(t,m,'LineWidth',1.5);
grid on;
title('Message Signal');
xlabel('Time (s)');
ylabel('Amplitude (V)');
subplot(2,1,2);
plot(t,fm_signal,'LineWidth',1);
grid on;
title('FM Signal');
```

```
xlabel('Time (s)');
ylabel('Amplitude (V)');
```

Output:



SAMPLE LAB EXAM QUESTIONS:

1. A communication engineer is testing an FM transmitter that will be used in a wireless voice communication system. During the design phase, the engineer needs to determine the maximum frequency deviation produced by the FM modulator to ensure that the transmitted signal remains within the allocated channel bandwidth. The FM modulator is configured with the following parameters: Message Signal.Amplitude (Am) = 2 V, Frequency Sensitivity (kf) = 1000 Hz/V. The engineer must verify the theoreticalfrequency deviation and confirm the result through MATLAB simulation before deploying the system.

Tasks:

1. Calculate the theoretical maximum frequency deviation (Δf\Delta fΔf) of the FM signal.

2. Develop a MATLAB program to generate the message signal and FM signal.

3. Compute the frequency deviation using the given FM parameters.

4. Display the calculated frequency deviation in the MATLAB Command Window.

5. Plot the message signal and the corresponding FM signal in the time domain.

Program:

clc; clear; close all;

% Parameters Am = 2; % Message amplitude (V) kf = 1000; % Frequency sensitivity (Hz/V) fm = 100; % Message frequency (Hz) Ac = 5; % Carrier amplitude (V) fc = 5000; % Carrier frequency (Hz) fs = 50000; % Sampling frequency (Hz)

% Calculate frequency deviation df = kf * Am; fprintf('Maximum Frequency Deviation = %.2f Hz\n', df);

% Time vector t = 0:1/fs:0.02;

% Message signal m = Am*cos(2*pi*fm*t);

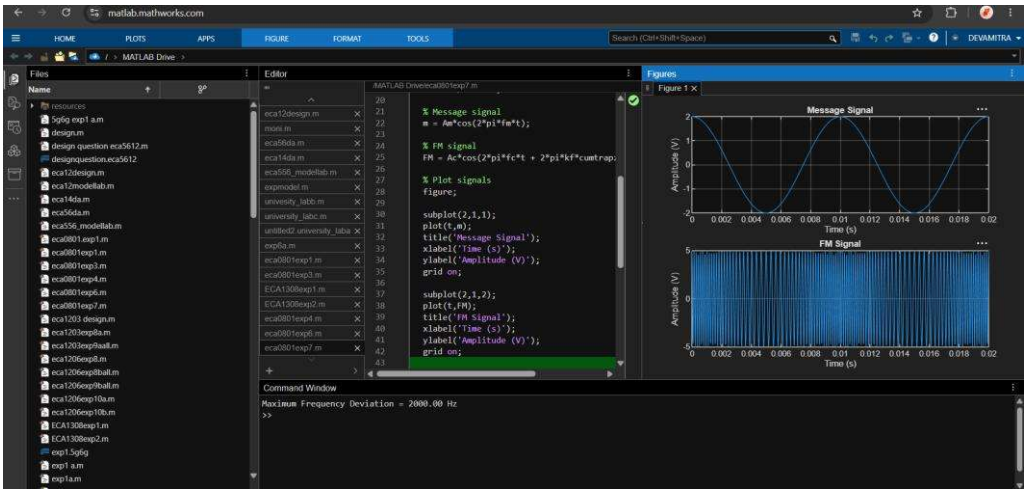
% FM signal FM = Ac*cos(2*pi*fc*t + 2*pi*kf*cumtrapz(t,m));

% Plot signals figure;

subplot(2,1,1); plot(t,m); title('Message Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;

subplot(2,1,2); plot(t,FM); title('FM Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;

Output:



2. A broadcast engineer is evaluating the performance of an FM transmitter before it is deployed in a local radio station. To ensure proper signal quality and bandwidth utilization, the engineer must determine the modulation index of the FM signal and verify the result through MATLAB simulation. During testing, the transmitter is found to produce a maximum frequency deviation of 2000 Hz when modulated by an audio signal of frequency 500 Hz. The engineer needs to calculate the modulation index and compare

the theoretical value with the value obtained from the MATLAB-generated FM signal. Given, Frequency Deviation (Δf) = 2000 Hz, Message Frequency (fm) = 500 Hz. Tasks

- Calculate the theoretical modulation index (β) of the FM signal.
- Develop a MATLAB program to generate the corresponding FM signal.
- Determine the modulation index using the simulation parameters.
- Display the calculated modulation index in the MATLAB Command Window.
- Plot the message signal and FM signal in the time domain.

Program:

clc; clear; close all;

% Parameters df = 2000; % Frequency deviation (Hz) fm = 500; % Message frequency (Hz) Am = 1; % Message amplitude (V) Ac = 5; % Carrier amplitude (V) fc = 10000; % Carrier frequency (Hz) fs = 100000; % Sampling frequency (Hz)

% Calculate modulation index beta = df/fm; fprintf('FM Modulation Index = %.2f\n', beta);

% Time vector t = 0:1/fs:0.01;

% Message signal m = Am*cos(2*pi*fm*t);

% FM signal FM = Ac*cos(2*pi*fc*t + beta*sin(2*pi*fm*t));

% Plot signals figure;

subplot(2,1,1); plot(t,m); title('Message Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;

subplot(2,1,2); plot(t,FM); title('FM Signal'); xlabel('Time (s)'); ylabel('Amplitude (V)'); grid on;

Output:

